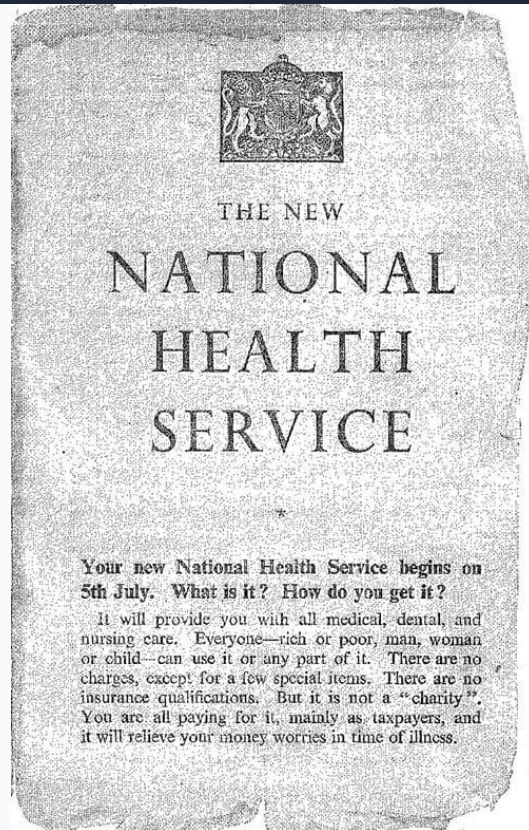




MEONCROSS SCHOOL | HISTORY DEPARTMENT

EDITION 2026.1

# WHY HAS MEDICINE ADVANCED MORE IN THE 20TH CENTURY THAN IN ALL PREVIOUS HISTORY COMBINED?

*Paper 1: Medicine Through Time with the Western Front – Modern*

Scholar: \_\_\_\_\_

Class: \_\_\_\_\_

### Edexcel Specification: c1900–present

- **1. Causes of disease:** Lifestyle factors, genetics, and the role of the NHS.
- **2. Prevention:** Immunisation programmes, screening, and public health campaigns.
- **3. Treatment:** Fleming and penicillin; blood transfusions; transplants; radiotherapy.
- **4. Modern Britain:** The founding of the NHS (1948) and its ongoing impact.

## PROGRESS & ASSESSMENT TRACKER

TARGET GRADE: \_\_\_\_\_

| Grade                                                                                | 9-1 GCSE Grading Scale |       |                  |
|--------------------------------------------------------------------------------------|------------------------|-------|------------------|
| Lesson / Assessment Title                                                            | Effort                 | Level | Teacher Comments |
| <b>L1: KT4.1: How have modern discoveries changed our understanding of illness?</b>  |                        |       |                  |
| <b>L2: KT4.2: How have prevention and treatment advanced in the modern era?</b>      |                        |       |                  |
| <b>L3: KT4.3: How was Penicillin discovered and mass-produced?</b>                   |                        |       |                  |
| <b>L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?</b> |                        |       |                  |
| <b>Final Unit Grade:</b>                                                             |                        |       |                  |

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# L1: KT4.1: How have modern discoveries changed our understanding of illness?

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LEARNING OBJECTIVES

☐ Explain how the discovery of DNA and the Human Genome Project transformed the understanding of genetic and hereditary illnesses.

☐ Analyze the influence of lifestyle choices on modern health and the impact of technology on medical diagnosis.

Do Now Activity

Score: / 10

What disease did Edward Jenner vaccinate against in 1796?

What did Edward Jenner observe that led to his vaccine?

Who discovered the cause of cholera in 1854?

Which Roman physician developed the Theory of Opposites?

What was the Royal Society?

Who introduced antiseptic surgery using carbolic acid in 1865?

What did Harvey prove wrong about Galen's theories of blood?

Who published the Germ Theory in 1861?

What invention helped spread new medical ideas during the Renaissance?

What was 'transference'?

Vocabulary Check

**The Odd One Out:** Circle ONE term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

DNA | Rosalind Franklin | Human Genome Project | Biopsy | Electron Microscope | Inherited Diseases

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KNOWLEDGE CHECK (Q1)

How did Watson and Crick's discovery of the double-helix structure of DNA help scientists search for the causes of hereditary diseases?

- ☐ A. It provided a complete blueprint of human biology, allowing scientists to identify specific faulty genes responsible for conditions like cystic fibrosis.
- ☐ B. It proved that hereditary diseases were caused by viruses.
- ☐ C. It allowed scientists to create antibiotics to kill hereditary diseases.
- ☐ D. It proved that hereditary diseases were caused by miasma.

Q2. Identify who discovered the double-helix structure of DNA in 1953.

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Q3. Describe the goal of the Human Genome Project.

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Q4. Explain why Germ Theory could not explain all illnesses.

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**Q5. Evaluate the impact of the discovery of DNA on modern medicine.**

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### Pair & Share Activity

**Q6. Prompt: Which modern development has had a more significant impact on our understanding of the causes of illness: the discovery of genetic factors (DNA and hereditary disease) or the scientific identification of lifestyle factors (smoking, diet, and alcohol)?**

*Think: Jot down your choice and one piece of evidence from the sources (e.g., how mapping the genome allows women to identify risks for breast cancer, versus how smoking is proven to be the biggest cause of preventable diseases in the world, causing heart disease and lung cancer).*

**Your Notes:**

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### Historian's Corner: The Debate over the Significance of DNA vs. Germ Theory

*GCSE historians frequently debate whether the 1953 discovery of the structure of DNA represents a more significant turning point in understanding illness than Pasteur's 1861 Germ Theory. Some historians argue that DNA is the ultimate breakthrough because it finally solved the mystery of non-communicable, hereditary conditions (such as Down's syndrome, haemophilia, and cystic fibrosis) that germs could never explain. However, other historians point out that while Germ Theory immediately transformed surgery, public health, and vaccination in the nineteenth century, the practical impact of DNA has been much slower to develop. Although we can map the genome, genetic treatments like gene therapy are still not widely available for most diseases, meaning the everyday clinical impact of genetics remains a future promise rather than a current reality.*

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**Q8. Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness.**

[12 marks • 15 mins]

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**Q9. Explain one way in which ideas about the cause of illness in the modern period (c1900-present) were different from ideas in the period c1700-c1900.**

[4 marks • 5 mins]

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**Q10. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c.1900-present.' How far do you agree? Explain your answer.**

**You may use the following in your answer:**

- Watson and Crick
- Lifestyle factors

**You must also use information of your own.**

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## General Notes

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## L2: KT4.2: How have prevention and treatment advanced in the modern era?

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### LEARNING OBJECTIVES

- ☐ Explain how scientific and technological developments led to the creation of magic bullets and antibiotics.
- ☐ Analyze the impact of the National Health Service (NHS) on the provision of and access to medical care.

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### Do Now Activity

Score:    / 10

What structure did Watson and Crick discover in 1953?

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What is the function of DNA in relation to disease?

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Name one modern lifestyle factor that causes disease.

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Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery?

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Why did the mapping of the human genome not lead to immediate cures?

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What did Harvey prove wrong about Galen's theories of blood?

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Who published the Germ Theory in 1861?

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Who introduced antiseptic surgery using carbolic acid in 1865?

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Who wrote 'On the Fabric of the Human Body' (1543)?

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What was the significance of the Public Health Act of 1875?

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Vocabulary Check

**The Golden Sentence:** Choose TWO terms from the word bank. Write ONE grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (because, although, or consequently):

Magic Bullet | Salvarsan 606 | Prontosil | Antibiotics | National Health Service (NHS) | Beveridge Report

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KNOWLEDGE CHECK (Q1)

What is the key scientific difference between an antiseptic like carbolic acid and a magic bullet like Salvarsan 606?

- ☐ A. Antiseptics are used on the outside of the body to prevent infection, whereas magic bullets are injected into the body to target specific microbes.
- ☐ B. Antiseptics are created from microorganisms, whereas magic bullets are synthetic chemicals.
- ☐ C. Antiseptics are used to cure viruses, whereas magic bullets cure bacteria.
- ☐ D. There is no difference; they are both the same thing.

Q2. Identify the first 'Magic Bullet' discovered in 1909.

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Q3. Describe how Magic Bullets work.

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Q4. Explain how diagnosing illness changed in the 20th century.

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**Q5. To what extent did the discovery of Magic Bullets revolutionize treatment?**

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### Pair & Share Activity

**Q6. Prompt: Which development was the most significant turning point in twentieth-century healthcare: the scientific breakthrough of antibiotics, or the political creation of the National Health Service (NHS)?**

*Think: Jot down your choice and one piece of evidence from the sources (e.g., how penicillin cured previously fatal infections, or how the NHS removed the fear of poverty by providing free healthcare for millions).*

**Your Notes:**

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### Historian's Corner: The Debate over the Significance of the Beveridge Report in the Foundation of the NHS

*Historians debate the extent to which the 1942 Beveridge Report was the primary catalyst for the creation of the NHS in 1948. Traditional consensus histories argue that William Beveridge's report, which recommended tackling the 'five giants' of want, disease, ignorance, squalor, and idleness 'from the cradle to the grave', created an irresistible democratic mandate that forced the post-war Labour government to act. However, revisionist political historians argue that the Beveridge Report was merely a set of recommendations and that the true turning point was the political tenacity of Health Minister Aneurin Bevan. They point out that Bevan had to wage a fierce political battle against the British Medical Association (BMA), who heavily resisted state control, and that the NHS was only realized because Bevan agreed to 'stuff their mouths with gold' by allowing consultants to keep their lucrative private practices alongside their NHS hospital roles.*



**Q7. Explain why the government took a more active role in preventing disease in the period c1900-present.**

[12 marks • 15 mins]

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**Q8. Explain one way in which ideas about the prevention of illness in the modern period (c1900-present) were different from the medieval period (c1250-c1500).**

[4 marks • 5 mins]

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**Q9. Explain why there was rapid progress in disease prevention in the period c1900-present.**

[12 marks • 15 mins]

**You may use the following in your answer:**

- Government vaccination campaigns
- Legislation

**You must also use information of your own.**

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## General Notes

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## L3: KT4.3: How was Penicillin discovered and mass-produced?

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### LEARNING OBJECTIVES

- ☐ Contrast the roles of Alexander Fleming with those of Howard Florey and Ernst Chain in the penicillin breakthrough.
- ☐ Analyze how the factors of government intervention, science and technology, and war collaborated to enable the mass production of penicillin.

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### Do Now Activity

Score:   / 10

What is a 'magic bullet'?

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Name the first magic bullet, discovered by Paul Ehrlich in 1909.

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What was established in Britain in 1948 to provide free medical care?

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Who was the Minister of Health responsible for establishing the NHS in 1948?

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Name the second magic bullet, discovered by Gerhard Domagk in 1932 to treat strep infections.

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What structure did Watson and Crick discover in 1953?

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Who published the Germ Theory in 1861?

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Who introduced antiseptic surgery using carbolic acid in 1865?

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Name one modern lifestyle factor that causes disease.

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Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery?

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Vocabulary Check

Conceptual Classification: Categorise the terms from the word bank into the two historical boxes below:

Penicillin | Alexander Fleming | Florey and Chain | Culture Medium | Mass Production | Antibiotic Resistance

Power, Governance & Warfare

Economy, Trade & Society

Q1. The Discovery and Mass Production of Penicillin

GCSE TWO-SIDED ARGUMENT

GCSE Analytical Assessment: Evaluate the miraculous antibacterial breakthrough of penicillin against its industrial challenges and modern limitations.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>▲ Antibiotic Miracle (Change)</div> <ul style="list-style-type: none"><li>Alexander Fleming observed in 1928 that penicillium notatum mold destroyed staphylococcus bacteria colonies on an open culture plate.</li><li>Howard Florey and Ernst Chain at Oxford (1938–1941) successfully isolated and purified penicillin, proving it cured deadly bacterial infections in mice and human trials.</li><li>US industrial scale and War Production Board funding produced 2.3 million doses in 1944, saving thousands of Allied soldiers on D-Day and eliminating pneumonia/gangrene.</li></ul> <p><i>Starter: “On the one hand, the development of penicillin was the greatest therapeutic breakthrough of the 20th century because...”</i></p> | <div>■ Industrial Fragility &amp; Antibiotic Resistance</div> <ul style="list-style-type: none"><li>Fleming abandoned his discovery in 1929 because he could not extract enough pure penicillin; without Florey, Chain, and US funding, it would have remained a laboratory footnote.</li><li>Oxford trials showed how fragile the supply was: policeman Albert Alexander showed rapid recovery but died when the team ran out of penicillin.</li><li>Penicillin was completely useless against viral infections like influenza, measles, or the common cold.</li><li>Modern over-prescription in humans and livestock has bred dangerous antibiotic-resistant superbugs like MRSA.</li></ul> <p><i>Counter: “However, the penicillin breakthrough had profound limitations because...”</i></p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

GCSE Paragraph Builder: Write a balanced GCSE exam paragraph evaluating how far penicillin was an individual discovery versus an industrial collaboration with modern vulnerabilities.

Connectives: Although Fleming made the initial serendipitous discovery... • Crucially, transforming this into a viable drug required... • Furthermore, in contemporary medicine... • Ultimately...

Q2. Describe how Penicillin was discovered.

KNOWLEDGE CHECK (Q3)

What was the main scientific and practical limitation that caused Fleming to abandon his work on penicillin in 1931?

- ☐ A. He realized that penicillin was highly toxic to humans.
- ☐ B. He could not secure funding from the government because it was peacetime.

- ☐ C. He lacked the chemical expertise and facilities to extract and purify penicillin in large enough quantities for human trials.
- ☐ D. He was drafted into the army to fight in World War II.

**Q4. Explain the roles of Florey and Chain in the development of Penicillin.**

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**Q5. Evaluate the role of World War II in the mass production of Penicillin.**

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### Pair & Share Activity

**Q6. Prompt: Who deserves the most credit for the penicillin breakthrough: Alexander Fleming for his accidental discovery in 1928, or Howard Florey and Ernst Chain for their scientific development and therapeutic isolation of the drug in the 1940s?**

*Think: Jot down your choice and one piece of evidence from the sources (e.g., Fleming recognizing and publishing the mould's bacteria-killing properties in 1928, or the Oxford team actually turning it into a usable drug and proving its clinical effectiveness on humans).*

**Your Notes:**

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**Historian's Corner: The Debate over the 'Myth' of Fleming and the Penicillin Breakthrough**

*For decades, popular historical accounts portrayed Alexander Fleming as the sole hero of the penicillin story, celebrating his accidental observation of the Penicillium mould in 1928 as an instant cure. However, revisionist historians actively challenge this traditional narrative, arguing that it propagates a 'myth'. They point out that Fleming actually abandoned his work on penicillin by 1931, having failed to isolate the active substance or prove that it could kill bacteria in living human blood. Revisionists argue that the true breakthrough belongs to Howard Florey, Ernst Chain, and the Oxford research team, who systematically isolated the drug in 1940 and proved its therapeutic value, but were initially ignored by a public captivated by the romantic narrative of Fleming's accident.*



**Q7. Explain why there was rapid progress in the development of penicillin in the 1940s.**

[12 marks • 15 mins]

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**Q8. Explain one way in which the treatment of illness in the modern period (c1900-present) was different from the treatment of illness in the period c1700-c1900.**

[4 marks • 5 mins]

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**Q9. 'Alexander Fleming was the most important individual in the development of penicillin.'**  
**How far do you agree? Explain your answer.**

[20 marks • 25 mins]

**You may use the following in your answer:**

- Staphylococcus
- Florey and Chain

**You must also use information of your own.**

- Staphylococcus
- Florey and Chain

**You must also use information of your own.**

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## General Notes

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## L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

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### LEARNING OBJECTIVES

- ☐ Explain how modern science and technology have transformed the diagnosis and treatment of lung cancer since c1900.
- ☐ Analyze the effectiveness and changing focus of government legislation to prevent smoking-related lung cancer.

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### Do Now Activity

Score:    / 10

Who discovered Penicillin by accident in 1928?

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Which two scientists at Oxford developed Penicillin into a usable drug?

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What major global event drove the mass production of Penicillin?

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What is a 'magic bullet'?

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What was established in Britain in 1948 to provide free medical care?

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Who was the Minister of Health responsible for establishing the NHS in 1948?

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What are the four humours?

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Who wrote 'On the Fabric of the Human Body' (1543)?

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Why is lung cancer so difficult to treat effectively compared to other diseases?

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Which landmark 1950 epidemiological study proved the link between smoking and lung cancer?

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Vocabulary Check

***Spot the Deliberate Error:** Read the statement below. One historical fact or vocabulary concept has been deliberately falsified. Underline the error and explain the accurate historical reality below:*

**Carcinogen | Bronchoscopy | CT Scanner | Radiotherapy | Immunotherapy | Nanny State**

*Challenge:* Write ONE statement containing a deliberate historical misconception using **Carcinogen** or **Bronchoscopy**. Swap with a partner to identify and correct the error:

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**Q1. Identify one method used to diagnose Lung Cancer.**

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**Q2. Describe the aggressive treatments used for Lung Cancer.**

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**Q3. Explain why the UK government has focused heavily on preventing Lung Cancer.**

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**KNOWLEDGE CHECK (Q4)**

**What is the key clinical advantage of using a CT scan over a traditional chest X-ray when diagnosing a lung cancer patient?**

- ☐ A. A CT scan uses harmless sound waves instead of radiation.
- ☐ B. A CT scan can be performed at home by the patient.

- ☐ C. A CT scan creates highly detailed, 3D cross-sectional images of the lungs, allowing doctors to detect much smaller tumors than a 2D X-ray.
- ☐ D. A CT scan can actively cure the lung cancer by destroying the tumor.

**Q5. Evaluate the effectiveness of government legislation in reducing smoking.**

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### Pair & Share Activity

**Q6. Prompt: Which category of modern government action has been the most significant turning point in the prevention of lung cancer: directly warning and educating current smokers (such as plain packaging and graphic warning labels), preventing young people from starting (such as raising the legal age of sale from 16 to 18), or protecting non-smokers from second-hand smoke (such as the 2007 indoor smoking ban)?**

*Think: Jot down your choice and one piece of evidence from your knowledge (for instance, the 2007 ban significantly shifting social attitudes about where smoking is acceptable, or how 1950s research first proved the definitive link between tobacco and cancer).*

**Your Notes:**

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### Historian's Corner: The Debate over the Motives for Modern Public Health Interventions

*Historians of modern medicine actively debate what primarily drove the British government to finally abandon its traditional hands-off approach and launch aggressive campaigns against smoking. Traditionalist historians argue that the state acted out of genuine humanitarian concern and scientific responsibility, moving to intervene once Richard Doll and A. Bradford Hill's landmark 1950 study provided undeniable statistical proof linking tobacco to lung cancer. Conversely, revisionist historians argue that the government's response was*

*actually incredibly slow and reluctant, delayed for decades by a desire to protect massive tobacco tax revenues and a fear of political backlash against state overreach. These historians suggest that the real turning point was economic pragmatism: as the financial burden of treating chronic, smoking-related illnesses on the newly established National Health Service (NHS) became unsustainable, the government was forced to recognize that prevention was far cheaper than clinical treatment.*



**Q7. Explain how science and technology have improved the diagnosis and treatment of lung cancer.**

[12 marks • 15 mins]

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**Q8. Explain one way in which hospital care in the modern period (c1900-present) was different from hospital care in the period c1700-c1900.**

[4 marks • 5 mins]

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**Q9. Explain why the government took action to improve public health in the period c1900-present.**

[12 marks • 15 mins]

**You may use the following in your answer:**

- Lung cancer
- The NHS

**You must also use information of your own.**

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## General Notes

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# End of Unit Reflection & Pupil Voice

*Unit: Paper 1: Medicine Through Time*

**Instructions:** Now that you have completed your final assessment, take 10 minutes to reflect on your learning across this entire unit. Be honest—this is your chance to use your **Pupil Voice** to help your teacher plan future lessons.

## 1. What Went Well (WWW)

What topic or skill did you find most engaging or easiest to understand in this unit?

## 2. Even Better If (EBI)

What did you find most challenging? Are there any concepts you are still confused about?

## 3. Effort & Target Setting

Circle your effort level for this unit: **1 (Low)**   **2**   **3**   **4**   **5 (Excellent)**

What is one specific target you want to set for yourself in the next unit?

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## Teacher Feedback & Coaching

(To be completed during live marking / feedback lessons)

## Factors Overview: Modern

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

| Factor                           | Specific Historical Example & Impact |
|----------------------------------|--------------------------------------|
| <b>Individuals</b>               |                                      |
| <b>The Church &amp; Religion</b> |                                      |
| <b>Government &amp; Wealth</b>   |                                      |
| <b>Science &amp; Technology</b>  |                                      |
| <b>Attitudes in Society</b>      |                                      |
| <b>War</b>                       |                                      |